Stochastic Processes 2021/22 Semester 1

Tutorial 5

1. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5,6\}$ and has transition matrix

$$\begin{pmatrix} 1/3 & 0 & 0 & 2/3 & 0 & 0 \\ 3/8 & 1/8 & 1/4 & 0 & 0 & 1/4 \\ 0 & 0 & 1/4 & 0 & 3/4 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/4 & 0 & 3/4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

   (a). Identify the closed classes.

   (b) Determine the limiting distribution if the Markov chain starts from the initial state $X_0 = 3$.

   (c) Determine the probability of absorption in state 6 if the Markov chain starts from the state $X_0 = 2$.

   (d) Determine the probability that the Markov chain hits the set $\{1,4\}$ if the Markov chain starts from the state $X_0 = 2$.

   (e) Determine the limiting distribution if the Markov chain starts from $X_0 = 2$.

2. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5,6\}$ and has transition matrix

$$\begin{pmatrix} 0 & 1/3 & 0 & 2/3 & 0 & 0 \\ 0 & 0 & 1/4 & 0 & 0 & 3/4 \\ 1/2 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 3/4 & 0 & 0 & 1/4 \\ 0 & 2/3 & 0 & 1/3 & 0 & 0 \\ 1/4 & 0 & 0 & 0 & 3/4 & 0 \end{pmatrix}$$

   (a) Classify the state space.

(b) Does there exist a limiting distribution if the Markov chain starts
(i) from the state $X_0 = 1$?
(ii) from the state $X_0 = 3$?
(iii) from a random initial value with distribution

$$\boldsymbol{\mu} = (1/4, 1/4, 0, 0, 0, 1/2)?$$

(iv) from a random initial value with distribution

$$\boldsymbol{\mu} = (0, 0, 0, 1/3, 1/3, 1/3)?$$

(v) from a random initial value with distribution

$$\boldsymbol{\mu} = (1/3, 0, 1/3, 0, 0, 1/3)?$$

(vi) from a random initial value with distribution

$$\boldsymbol{\mu} = (1/6, 2/9, 0, 1/9, 1/6, 1/3)?$$

Provide your reasoning.

3. Let $\{X_n\}$ be a Markov chain with the state space $S = \{1, 2, 3, 4\}$ and transition matrix

$$P = \begin{pmatrix} 0 & 1/3 & 0 & 2/3 \\ 1/3 & 0 & 2/3 & 0 \\ 0 & 3/4 & 0 & 1/4 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}$$

(i) Find its stationary distribution. You may use both ways (direct and by reduction).
(ii) For which initial distributions does the limiting distribution exist?

4. Consider the Markov chain from *Tutorial 3, Q5(c)*.
(i) Determine the family of all stationary distributions for this Markov chain.
(ii) What is the limiting distribution for this Markov chain if it starts from $X_0 = 2$?